



Polluting resource extraction and climate risk

Israa Hashem¹ | Walid Marrouch^{2,3} 

¹Department of Economics, Concordia University, Montreal, Quebec, Canada

²Department of Economics, Lebanese American University, Beirut, Lebanon

³CIRANO, Montreal, Quebec, Canada

Correspondence

Walid Marrouch, Department of Economics, Lebanese American University, P.O. Box: 13-5053 Chouran, Beirut 1102 2801, Lebanon.

Email: walid.marrouch@lau.edu.lb

Abstract

Using a fossil fuel extraction model that treats the atmosphere as a depletable resource, we study the optimal price of carbon in the presence of endogenous uncertainty around a climatic regime shift. We find that the optimal carbon tax should account an uncertainty-adjusted cost term associated with the environment's scarcity. This term is shown to be sensitive to the natural sequestration rate of the atmosphere and to the probability surrounding a climate tail event. Our analysis also shows that in the presence of uncertainty, the shadow price of the environment should grow at a faster rate. Lastly, compared to the endogenous uncertainty case, we find that if the probability surrounding a regime shift is exogenously given, this shadow price should even grow at a higher rate.

KEYWORDS

climate risk, fossil fuel, scarcity, uncertainty

Considerations for Management

Based on our theoretical analysis, considerations for resource management may include:

- Beside direct damages, optimal carbon pricing should account for a cost term related to the environment's scarcity.

This is an open access article under the terms of the Creative Commons Attribution License, which permits use, distribution and reproduction in any medium, provided the original work is properly cited.

© 2022 The Authors. *Natural Resource Modeling* published by Wiley Periodicals LLC.



- As the probability of facing an extreme climate event increase, this cost term amplifies.
- The cost term disappears for a sufficiently high abatement effort.
- Compared to endogenous uncertainty, exogenous uncertainty enforces placing a higher value on the environment's shadow price.

1 | INTRODUCTION

Over the past decades, it has become increasingly understood that the environment has its own limitations. Boulding (1966) alluded to an economy that is tied to limited reservoirs whether of resources or as pollution sinks. Likewise, Simpson et al. (2005) introduces the *New Scarcity* which he defines as the limitations of the environment as a pollution sink. In the specific context of climate change caused by anthropogenic emissions of greenhouse gases, this definition also applies to carbon sinks. Discernibly, the atmosphere, as one facet of the environment, is limited in its capacity to infinitely accommodate a growing trend in the concentration of accumulated carbon dioxide (CO₂) emissions. Humanity already started facing global warming mostly due to the excess of atmospheric CO₂.¹ In addition to the direct damages of using fossil fuels, apparently, there is an indirect loss in the process, which is the scarcity cost of clean atmosphere that is no longer available for future generations. The unavailability of the sink stock in the future stems from the fact that CO₂ would stay for hundreds of years in the atmosphere with higher potential devastating impacts of growing concentrations.²

In this short paper, we introduce an extraction model of a polluting nonrenewable resource, mainly fossil fuels, while incorporating the scarcity of the atmosphere as a pollution sink. We will refer to fossil fuels as the resource and to clean atmosphere as the environment. Our model's formulation allows us to differentiate between two types of damages due to the continuous accumulation of CO₂ in the atmosphere. First, there is the direct damage that is associated with the global warming effect (or the stock effect as referred to in the literature), which is featured using the damage function. Second, the indirect damage is tied to the loss of functions of the environment (or the opportunity cost of using up the atmosphere as a pollution sink) that may become suddenly unavailable for future generations. By modeling the atmosphere as a depletable resource (instead of only using a damage function that depends on the pollution stock), we can capture the environment's scarcity. Also, this approach allows us to explicitly analyze the above-mentioned indirect cost in the presence of uncertainty around a sudden loss in the functions of the environment. In this framework, we first analyze the optimal carbon tax components in light of the potential externality cost due to the environment's scarcity. The other objective is to explore the implications of introducing endogenous uncertainty surrounding the realization of climate tail events. By climate tail events, or a regime shift as referred to in the literature, we mean the incidence of a sudden discrete jump in damages, that may cause irreversible loss in the functions of the environment, which in turn may leave severe impact for life on earth. For instance, Nordhaus (2011) discusses



the possibility of potential catastrophic declines in consumption as a result of the temperature response to surged accumulations of greenhouse gases. The possibility of a shift in environmental damage has been also examined in the context of environmental agreements (as in Nkuiya et al., 2015). However, uncertainty of a regime shift in a polluting resource extraction model, that also treats the atmosphere as a scarce resource, has not been modeled in the literature. We, thus, model a tail event that is directly related to the environment's scarcity to fill this gap in the literature.

A range of precursory research has tackled optimal extraction and environmental externalities. Chakravorty et al. (2008) looked at the sequence of extraction and the carbon tax in the presence of two polluting nonrenewable resources. Chakravorty et al. (2006) examine a Hotelling model constrained by a ceiling on the stock of pollution and explain how it affects the dynamic transition to a renewable resource. Other models of nonrenewable resources with a clean backstop investigate the optimal trajectory of the carbon tax, and find that the tax depends on how the carbon stock would evolve over time (Hoel & Kverndokk, 1996; Tahvonen, 1997). Sandal and Steinshamn (2008) model the effect of nonlinear pollution decay on the optimal extraction of nonrenewable resources. Farzin (1996) studies the environmental stock externality with a threshold on the pollution stock. In our model, we refrain from adding an additional clean energy resource since the objective of this study is to examine the cost of carbon in light of uncertainty around a catastrophic climate event and not to conduct a dynamic analysis (or optimal regime transition). Accounting for a backstop would be reflected in the benefit function and the cost function associated with utilizing the resource (see Tahvonen, 1997). As such, this will result in an augmented expression of the full social cost of carbon and will also modify the dynamics governing the polluting resource extraction. However, our model proposes general results using the necessary conditions solely and without introducing specific functional forms or extending the discussion to tackle dynamic analyses. Thus, we expect ignoring an alternative energy source would not qualitatively affect our proposed results. Moreover, all these models employ a damage function that depends on the pollution stock, yet none depicts the environment's scarcity by explicitly featuring the atmosphere as a resource. Thus, our work contributes to this literature by studying optimal carbon pricing under the environment's scarcity in the presence of endogenous uncertainty of climate tail events.

We arrive at the following results which can be fundamental to the theory of carbon pricing. First, we emphasize the result obtained by Farzin (1996) stating that the optimal carbon tax should account for a cost term related to the environment's scarcity. However, unlike in Farzin (1996) where this cost term is shown as a shadow price, we find an uncertainty-adjusted term that encompasses the pollution shadow price.³

We further show that this term is sensitive to the natural sequestration rate of the atmosphere and to the probability surrounding a climate tail event, and highly depends on the abatement effort. Second, our analysis shows that in the presence of uncertainty around a regime shift, the shadow price associated with the environment should grow at a faster rate. Lastly, compared to endogenous uncertainty, this cost should even grow at a higher rate if the probability surrounding a regime shift is exogenously given.

Next, we present our model and results in Section 2 followed by Section 3, which concludes.



2 | THE MODEL

We present our social planner problem as follows. Let $S(t)$ denote the stock of fossil fuel at time t where $y(t)$ represents the resource extraction rate. The stock exists initially with certainty of size S_0 . Hence, Equation (1) governs the resource stock dynamics:

$$\dot{S}(t) = -y(t). \quad (1)$$

$B(y(t))$ is the benefit for society per extracted unit. The corresponding extraction costs function is $C_1(y(t), x(t))$, where $x(t)$ denotes the cumulative extraction to date and is given by: $x(t) = \int_0^t y(\tau) d\tau = S_0 - S(t)$. Let $E(t)$ denote the cumulative unabsorbed emissions at time t and $e(y(t))$ the net emissions where the latter is specified as: $\alpha(g(t))y(t)$. We define $\alpha(g(t))$ as the pollution transformation function, which is determined by the abatement effort, $g(t)$, such that: $\frac{d\alpha(g)}{dg} < 0$, $\alpha(g(t)) \in [0, 1]$ and $\alpha(0) = 1$. Moreover, $\rho(A(t)) \in [0, 1]$ and $A(t)$ denote the instantaneous natural regeneration rate of the atmosphere and the stock of clean atmosphere respectively; $\frac{d\rho}{dA} > 0$ since a cleaner atmosphere stock has a better capacity of regeneration.⁴ While the direct environmental damage function is perfectly known, extracting this resource is however associated with uncertainty caused by a sudden environmental hazard; this is referred to in the literature as a regime shift.⁵ One way to capture this feature is to treat ρ as a random variable that follows a binomial distribution. ρ exhibits a discrete jump and takes the value 0 with probability $P(A)$ or stays a continuous function of A with probability $[1 - P(A)]$. The former case would refer the regime shift state.⁶ Given that the sudden catastrophic event is tied to the function-loss of the environment, we model the regime shift as sudden sharp drop in the atmospheric regeneration rate, instead of reflecting this in the damage function.⁷ In this formulation, uncertainty is endogenous; a cleaner environment will be associated with a lower such probability, $\frac{dP}{dA} < 0$. If $\frac{dP}{dA} = 0$, it boils down to the exogenous uncertainty case. Whence, the evolution of cumulative net emission and the stock of clean atmosphere over time are interconnectedly given in Equations (2) and (3), respectively:

$$E(t) = \int_0^t [1 - (1 - P(A))\rho(A) - P(A) \cdot 0] \alpha(g(\tau))y(\tau) d\tau. \quad (2)$$

$$\dot{A}(t) = -\dot{E}(t) = -[1 - (1 - P(A))\rho(A)] \alpha(g(t))y(t). \quad (3)$$

Equation (2) points out that the cumulative unabsorbed emissions at a given period are the total of the fractions of the transformed pollution emission units, that are not sequestered, up to date. Similarly, the law of motion corresponding to the change of the stock of A shows that at each point in time, the stock of clean atmosphere is reduced by the quantity of emissions remaining net of natural sequestration and the usage of an available abatement technology.

Furthermore, let $C_2(g(t))$ denote the carbon emissions abatement cost and $D(E(e(y(t))))$ the environmental damage function. To ensure we have a maximum, the following standard assumptions from the resource literature are introduced: $\frac{dB}{dy} > 0$, $\frac{d^2B}{dy^2} \leq 0$, $\frac{\partial C_1}{\partial y} > 0$, $\frac{\partial^2 C_1}{\partial y^2} \geq 0$, and $\frac{\partial C_1}{\partial x} \geq 0$, indicating higher extraction costs as the resource stock gets depleted (as in Gaudet,



2007). We also assume the following: $\frac{dC_2(g)}{dg} > 0$, $\frac{d^2C_2(g)}{d^2g} \geq 0$, $\frac{dD(e)}{de} > 0$, $\frac{d^2D(e)}{d^2e} \geq 0$, $\frac{dD}{dE} > 0$ with $\frac{dE}{de} > 0$, and $\frac{de}{dy} > 0$.

The social planner wishes to allocate the extraction rate $y(t)$ and the abatement level $g(t)$ that maximize the total discounted net benefits from extraction over the period $[0, T]$, where T is the terminal date.⁸ The optimal control problem becomes:

$$\text{Max} \int_0^T [B(y(t)) - C_1(y(t), x(t)) - C_2(g(t)) - D(E(e(y(t))))] \exp(-rt) dt,$$

subject to $\dot{S}(t) = -y(t)$, $S(0) = S_0$, $\dot{A}(t) = -[1 - (1 - P(A))\rho(A)]\alpha(g(t))y(t)$, and $A(0) = A_0$,

where r denotes the social rate of discount.

The current value Hamiltonian is:

$$H_c = B(y(t)) - C_1(y(t), x(t)) - C_2(g(t)) - D(E(\alpha(g(t))y(t))) - \lambda(t)y(t) - \mu(t)[1 - (1 - P(A))\rho(A)]\alpha(g(t))y(t) \quad (4)$$

and the corresponding static necessary conditions are:

$$-\frac{dC_2}{dg(t)} - y(t)\frac{d\alpha(g(t))}{dg(t)}\left[\frac{dD}{dE}\frac{dE}{de} + \mu(t)[1 - (1 - P(A))\rho(A)]\right] = 0, \quad (5)$$

$$\frac{dB}{dy} - \left[\frac{\partial C_1}{\partial y} + \frac{\partial C_1}{\partial x}\frac{dx}{dy}\right] - \frac{dD}{dE}\frac{dE}{de}\alpha(g(t)) - \lambda(t) - \mu(t)[1 - (1 - P(A))\rho(A)]\alpha(g(t)) = 0. \quad (6)$$

$\lambda(t)$ and $\mu(t)$ denote the shadow prices of the resource and the environment respectively. Equation (5) governs the condition on choosing the optimal abatement level. It is inferred that increasing the abatement effort by a unit has a higher marginal benefit compared to the case of not accounting for the environment's scarcity. Hence, a more precise cost-benefit analysis carefully entails accounting for an additional welfare gain associated with a slower depletion of the atmosphere stock. Specifically, it has positive implications on the endogenous variables $\rho(A)$ and $P(A)$. A cleaner environment because of increasing abatement allows for: precluding an additional negative effect on the ability of the atmosphere to absorb emissions as well as preventing an additional increase in the probability of observing climatic tail events.

The other condition, given by Equation (6), insures that at the margin, the benefit from extracting fossil fuel is the same as the full marginal cost. The marginal cost includes the costs of extraction, abatement, and fossil fuel shadow price which are classical in the literature. The relatively new portion is the term associated with the scarcity of the environment. Alternatively, our first result, which is given by proposition 1, suggests that the optimal carbon tax is found at the intersection of the net marginal benefit and the full marginal damage of extraction.

Proposition 1. *The full social cost of carbon is the sum of explicit environmental damages caused by global warming and the shadow prices of the resource and of the environment.*



The full environmental cost of carbon (i.e., the full marginal damage) of each extracted unit has three components. $\frac{dD}{dE} \frac{dE}{de} \alpha(g(t))$ is the direct cost of the accumulated emissions, which contributes to global warming. The second term is the shadow price of the resource ($\lambda(t)$) while the last term is the uncertainty-adjusted shadow price of the environment ($\mu(t)[1 - (1 - P(A))\rho(A)]\alpha(g(t))$). The last two terms correspond to the optimal tax in the decentralized model of Sandal and Steinshamn (2008). Nevertheless, our centralized model picks up an additional term. It also differs by explicitly modeling the environment as a partially-depletable resource that is subject to climate risk. Farzin (1996) also shows that the stock effect results in a shadow cost and that the optimal carbon tax should account for it. Overall, we obtain a more general result associated with the environment's scarcity by also introducing the natural sequestration rate and uncertainty around its value. These new insights are presented in Proposition 2.

Proposition 2. *The optimal carbon tax should respect the following conditions regarding the cost associated with the environment's scarcity:*

- If $P(A) > 0$ or $\rho \neq 0$, the tax policy should account for the cost term associated with the environment's scarcity.
- For $P(A) = 1$ and $\rho(A) = 0$, the underlying cost term renders its highest possible value.
- This cost term disappears in two cases:
 - $P(A) = 0$ and $\rho(A) = 1$.
 - for a sufficiently high abatement effort.

Proposition 2 indicates that the cost associated with the scarcity of the environment is sensitive to the probability of a regime shift and to the natural sequestration rate of the atmosphere. If the probability is positive, the carbon tax should optimally account for the cost term associated with the environment's scarcity even when the natural sequestration rate takes its maximum value, $\rho = 1$. Similarly, if $\rho(A) \neq 0$, this term is strictly positive regardless of the probability value. Besides, a high (low) probability of witnessing a climatic tail event amplifies (reduces) the magnitude of this cost. However, it can be deduced that for a sufficiently high abatement effort, in the case of $\alpha(g(t))$ approaches 0, the optimal carbon policy need not incorporate this cost. This cost term also goes to zero if these two conditions hold simultaneously: $P(A) = 0$ and $\rho(A) = 1$; however, this case departs from reality as it means the environment is not scarce. Overall, a social cost of carbon that is composed of the costs of direct damages and resource scarcity acts as a lower bound.

The dynamic optimality conditions are given by Equations (7) and (8).

$$\dot{\lambda}(t) = r\lambda(t) - \frac{\partial H_c}{\partial S} = r\lambda(t) - \left[-\frac{\partial C_1}{\partial x} \frac{dx}{dS} \right] = r\lambda(t) - \frac{\partial C_1}{\partial x}, \quad (7)$$

$$\dot{\mu}(t) = r\mu(t) - \frac{\partial H_c}{\partial A} = r\mu(t) + \mu(t)\alpha(g(t))y(t) \left[\frac{dP}{dA}\rho(A) - (1 - P(A))\frac{d\rho}{dA} \right]. \quad (8)$$

Consistent with the literature, the dynamic condition on the resource's shadow price is standard. In the absence of extraction costs, Equation (7) is nothing but the pure Hotelling's rule.⁹ Of more interest to this paper is exploring the implications on the dynamic condition of



the environment's shadow price in the presence of endogenous uncertainty. To accomplish this, it first requires stating the following observation.

Observation. The dynamics governing the shadow price of the environment compared to that of a polluting resource exhibit an opposite interpretation. In the case of a polluting nonrenewable resource, a higher growth rate of its shadow price implies that a faster depletion should take place over time. By contrast, for the environment, it implies that the stock of clean atmosphere has appreciated in value and society should use it at a slower rate.

This observation stems from the fact that the environment is rather a clean resource, that is, a sink of the polluting resource. This allows us to state the next result.

Proposition 3. *If $P(A) > 0$, that is, in the presence of uncertainty of climate tail events, the shadow price of the environment grows at a faster rate.*

Proposition 3 is obtained from condition (8). The possibility of facing a regime shift gives rise to a higher growth rate of the environment's shadow price compared to the case where this probability is not prompted in the model. Also, having an endogenous sequestration rate plays a necessary role as to why this effect appears in the equation. Intuitively, when the environment's ability to regenerate itself depends on its state, a positive probability of a regime shift should play a role in increasing its value. A key policy implication of this result is that the likelihood of such tail events to happen compels conservationist steps aimed at protecting the environment.

Proposition 4. *Exogenous probability of a regime shift should enforce placing a higher value on the environment compared to the case when uncertainty is endogenous.*

The finding in proposition 4 is also deduced from Equation (8). We find that exogenous uncertainty leads to even more conservationist usage of the environment and thus higher growth of its shadow value. If the probability to observe a tail event is exogenously given, that is, $\frac{dP}{dA} = 0$, the growth rate of the environment scarcity price is higher. However, when this uncertainty is endogenous, it must be that $\frac{dP}{dA} < 0$ inducing the above result. One plausible interpretation of this finding is that when uncertainty is exogenous, society should be more cautious since there is no control over ρ through $A(t)$.

3 | CONCLUSION

In this paper, we examined the effect of treating the environment as a scarce resource and the implications of introducing uncertainty surrounding the possibility of a regime shift. To do so, we introduced an extraction model that depicts the atmosphere as a depletable resource. This model formulation allows us to explicitly differentiate between the indirect damage associated with the environment's scarcity and the direct environmental damage, which is represented by the damage function. Also, this approach allowed us to study the full social cost of carbon in the presence of endogenous uncertainty surrounding a tail event in the context of global warming. Our results show



that the cost related to the environment's scarcity is sensitive to the atmospheric sequestration rate, to the probability of observing a regime shift, and to the abatement level. We also find that in the presence of uncertainty, the shadow price of the environment should grow at a faster rate compared to the case when there is no uncertainty. Furthermore, compared to the endogenous uncertainty case, our results suggest that exogenous uncertainty enforces a more conservationist environmental rule. Under the discussed conditions, the externality cost term induced by the environmental scarcity should be a significant part of the social cost of carbon which turns out to be a lower bound as currently defined in the extant literature.

AUTHOR CONTRIBUTIONS

Israa Hashem: Conceptualization (equal); formal analysis (equal); investigation (equal); methodology (equal); project administration (equal); resources (equal); validation (equal); writing—original draft (equal); writing—review & editing (equal). **Walid Marrouch:** Conceptualization (equal); formal analysis (equal); methodology (equal); project administration (equal); resources (equal); validation (equal); writing – original draft (equal); writing – review & editing (equal).

CONFLICT OF INTEREST

The authors declare no conflict of interest.

DATA AVAILABILITY STATEMENT

Data sharing is not applicable to this article as no new data were created or analyzed in this study.

ORCID

Walid Marrouch  <http://orcid.org/0000-0001-8488-6555>

ENDNOTES

- ¹ A report by IPCC (2019) states that the global mean surface temperature (GMST) has risen, in the period 2006–2015, by around 0.87°C more than the average during 1850–1900. The report can be found at: <https://www.ipcc.ch/srccl/chapter/summary-for-policymakers/> While we acclaim acknowledging the environmental sinks scarcity, we do not make any stands on the extent to which warming has been caused by anthropogenic emissions of greenhouse gases. This is because there is still high controversy on climate sensitivity, which refers to the global temperature response to the growing atmospheric CO₂ concentration.
- ² See the IPCC fifth assessment report which can be found at: <https://www.ipcc.ch/report/ar5/wg2/>
- ³ In case of a complete transition to using backstop technology, we would expect this cost to be negligible. Otherwise, in the most likely case of coexistence of the polluting resource and a clean backstop, the carbon tax would account for an additional backstop term.
- ⁴ This representation is different from a similar work, Hoel and Kverndokk (1996), which treats the sequestration rate as a carbon depreciation parameter.
- ⁵ See Nkuiya et al. (2015), for an analogous approach to study the effects of introducing the regime-shift uncertainty on international environmental agreements.
- ⁶ The likelihood to face extreme-level events is widely considered in the literature on climate change, which also use the term tail events. In this framework, it is worth to mention the dismal theorem proposed by Martin Weitzman. The theorem states that standard economic analysis does not apply under limited conditions on the structure of uncertainty and preferences, where in tail events society has a large expected loss (Nordhaus, 2011). Nordhaus (2011) examines the dismal theorem on tail events in the context of climate change. We invite the reader to refer to this literature to avoid ambiguity on what is conveyed by a regime shift in this model.



- ⁷ While one may argue that the atmospheric sequestration of CO₂ is already negligible, it is worth noting that our model also captures the very long run time aspects of atmospheric CO₂.
- ⁸ If we were to solve for a specific solution, T would be endogenously determined as part of the solution; however, we do not focus on the extraction period as we aim to solve the model in its general form and interpret the implication of introducing endogenous uncertainty.
- ⁹ This rule, introduced by Hotelling (1931), requires the nonrenewable resource value in situ to rise at the rate of interest. Nevertheless, in our model, the fossil fuel shadow price rises at a rate lower than the rate of discount due to the presence of extraction costs stock effect, as found in the resource literature (see e.g.: Krautkraemer, 1998). Gaudet (2007) uses the physical assets equilibrium condition as a basis in his careful analysis of reconciling Hotelling's rule with empirical regularities. He also introduces modified versions of the rule upon accounting for factors such as varying costs and market structure.

REFERENCES

- Boulding, K. E. (1966). The economics of the coming spaceship earth. In H. Jarrett (Ed.), *Environmental quality in a growing economy*. Johns Hopkins University Press.
- Chakravorty, U., Magne, B., & Moreaux, M. (2006). A hotelling model with a ceiling on the stock of pollution. *Journal of Economic Dynamics and Control*, 30(12), 2875–2904.
- Chakravorty, U., Moreaux, M., & Tidball, M. (2008). Ordering the extraction of polluting nonrenewable resources. *American Economic Review*, 98(3), 1128–1144.
- Farzin, Y. H. (1996). Optimal pricing of environmental and natural resource use with stock externalities. *Journal of Public Economics*, 62(1–2), 31–57.
- Gaudet, G. (2007). Natural resource economics under the rule of hotelling. *Canadian Journal of Economics*, 40(4), 1033–1059.
- Hoel, M., & Kverndokk, S. (1996). Depletion of fossil fuels and the impacts of global warming. *Resource and Energy Economics*, 18(2), 115–136.
- Hotelling, H. (1931). The economics of exhaustible resources. *Journal of Political Economy*, 39(2), 137–175.
- IPCC. (2019). Summary for Policymakers. In P. R. Shukla, J. Skea, E. Calvo Buendia, V. Masson-Delmotte, H.-O. Pörtner, D. C. Roberts, P. Zhai, R. Slade, S. Connors, R. van Diemen, M. Ferrat, E. Haughey, S. Luz, S. Neogi, M. Pathak, J. Petzold, J. Portugal Pereira, P. Vyas, E. Huntley, K. Kissick, M. Belkacemi, & J. Malley, (eds.), *Climate change and land: An IPCC special report on climate change, desertification, land degradation, sustainable land management, food security, and greenhouse gas fluxes in terrestrial ecosystems*.
- Krautkraemer, J. A. (1998). Nonrenewable resource scarcity. *Journal of Economic Literature*, 36(4), 2065–2107.
- Nkuiya, B., Marrouch, W., & Bahel, E. (2015). International environmental agreements under endogenous uncertainty. *Journal of Public Economic Theory*, 17(5), 752–772.
- Nordhaus, W. D. (2011). The economics of tail events with an application to climate change. *Review of Environmental Economics and Policy*, 5(2), 240–257.
- Sandal, L. K., & Steinshamn, S. I. (2008). Optimal environmental taxes: Effects of pollution decay and consumer awareness. *Natural Resource Modeling*, 21(3), 436–465.
- Simpson, R. D., Toman, M. A., & Ayres, R. U. (2005). *Scarcity and growth revisited*. Natural resources and the environment in the new millenium. Resources for the Future.
- Tahvonen, O. (1997). Fossil fuels, stock externalities, and backstop technology. *Canadian Journal of Economics*, 30(4), 855.
- Withagen, C. (1994). Pollution and exhaustibility of fossil fuels. *Resource and Energy Economics*, 16(3), 235–242.

How to cite this article: Hashem, I., Marrouch, W. (2023). Polluting resource extraction and climate risk. *Natural Resource Modeling*, 36, e12354.

<https://doi.org/10.1111/nrm.12354>